



University of Tartu

Practical work 8: Advanced SQL (Aviation Edition)

Exercise 1: View Creation

Run the script (from Moodle) to create the Aircraft, Pilot, Base, and Mission tables in Postgres.

A. View creation

1. Create view `crew_manifest` allowing the operations officer to verify identities (pilot full name, role, base, clearance). Check view content and drop it.
2. Create `captain_high_hours` view on the Pilot table grouping pilots whose `flight_hours` are higher than the overall average and whose role is 'Captain'. Check results.
3. Create `young_28` view on the Pilot table grouping female pilots whose age is less than 28. The insertion or update on this view must always respect this constraint.
4. Set to 26 the age of the pilots in `young_28` whose first name is 'Amina'.
5. Create `mission_crew` view containing the details of the pilots assigned to mission 'Operation Night Hawk'.
6. Add a new female pilot with age < 27 into Pilot. Verify the view maintainability on `young_28`.

Exercise 2: Queries

B. Queries

1. List distinct aircraft categories (e.g., 'Fighter', 'Transport', 'Recon'). No duplicates.
2. Which pilots are stationed at a base whose name ends with 'Field' or 'AFB'? Sort by age descending.
3. Which pilots have flown missions commanded by 'Jean Dupont'? Sort by base name.
4. What are the minimum and maximum ages of pilots?
5. Which pilots are neither 'Captain', nor 'First Officer', nor 'Ground Crew'? Write the query using subqueries.
6. Who is the youngest pilot in the database? Display name, age, gender, and `flight_hours`.
7. What is the average `flight_hours` for pilots with role 'Captain'?
8. Compute, for each role, the number of pilots and the total `flight_hours`.
9. Modify the previous query to also break down results by gender within each role.

10. For each mission, compute the number of assigned pilots; also display mission code name and the mission commander's name.
11. For each mission, compute the number of female pilots assigned.

Exercise 3: More Queries

C. More queries

1. Show the average flight_hours for each base with at least 20 pilots; order by average flight_hours descending.
2. Find pilots whose flight_hours are higher than the average flight_hours of pilots at their own base.
3. List pilots stationed in NV or TX.
4. Find pilots not assigned to any mission.
5. Find pilots whose flight_hours are above the overall company-wide average.
6. Find pilots who have flown at least one mission commanded by 'Akari Ito'.
7. Show 5 pilot call signs in uppercase and the length of their email addresses.
8. Rank pilots within each base by flight_hours, highest hours rank 1.
9. Find pilots whose email ends with '@airops.gov'.